KNN & SVM — Interview Revision Q&A;

K-Nearest Neighbors (KNN)

Q1. How does KNN make a prediction?

KNN finds the K nearest training examples to a new point. For classification, it predicts the majority class among those neighbors. For regression, it typically averages their target values.

Q2. Why is feature scaling important in KNN?

KNN relies on distances. A feature with a much larger numerical range can dominate the distance calculation, making smaller-scale features contribute very little. Scaling puts features on comparable ranges.

Q3. What happens with very small vs. very large K?

Small K (such as K=1) gives low bias but high variance: predictions are sensitive to noise and outliers and can overfit. Large K produces smoother predictions but can wash out local patterns, increasing bias and causing underfitting.

Q4. What are KNN's main advantages and disadvantages?

Advantages: almost no explicit training, simple, and no learned weight parameters. Disadvantages: high storage requirements and potentially expensive prediction because distances may need to be computed against many training examples. KNN can generalize to unseen points if the training data is representative.

Key KNN takeaway:

KNN has almost no training cost but potentially high prediction cost.

Support Vector Machine (SVM)

Q1. What is SVM and its main objective?

SVM finds a decision boundary (hyperplane) that separates classes while maximizing the margin between the boundary and the closest training points.

Q2. What are support vectors?

Support vectors are the training points closest to the decision boundary. In the standard hard-margin formulation, they lie on the margin boundaries. They are important because they determine the position of the optimal hyperplane.

Q3. What is the margin and why maximize it?

The margin is the distance from the decision boundary to the closest points of the classes. A larger margin generally provides better robustness and generalization to unseen data. For the standard formulation, the full margin width is $2/||w||$.

Q4. Hard margin vs. soft margin?

Hard-margin SVM requires all training points to be correctly classified with the required margin, making it sensitive to noise and outliers. Soft-margin SVM allows margin violations and even some misclassifications, making it more robust.

Q5. What does C control?

C controls how strongly margin violations (hinge loss) are penalized. Large C heavily penalizes violations and approaches hard-margin behavior. Small C allows more violations and favors a wider margin/more regularization.

Q6. What is hinge loss?

For $y \in \{-1, +1\}$, hinge loss is $L = \max(0, 1 - y f(x))$. If a point is correctly classified and outside the margin, the loss is zero. Correct points inside the margin have positive loss, and misclassified points have positive loss.

Q7. Why use kernels?

Some data is not linearly separable in the original feature space. A nonlinear feature mapping can make it linearly separable in a higher-dimensional space. The kernel trick computes the required inner products in that space without explicitly constructing the transformed features.

Kernel trick:

$K(x,z) = \phi(x)^T\phi(z)$. Instead of explicitly calculating $\phi(x)$, SVM directly evaluates $K(x,z)$. For RBF, $K(x,z) = \exp(-\gamma||x-z||^2)$.

Q8. Common kernels and RBF gamma?

Common kernels include linear, polynomial, and RBF. In an RBF kernel, gamma controls how quickly similarity decays with distance. Large gamma gives very local influence and a more complex boundary; small gamma gives broader influence and a smoother boundary.

Q9. Why is feature scaling important for SVM?

SVM geometry relies on distances and dot products. A feature with a much larger scale can dominate these calculations and disproportionately affect the decision boundary. Scaling features prevents this.

Q10. Why can kernel SVM struggle with huge datasets?

Kernel SVMs often require pairwise kernel computations between training examples. The kernel matrix has size n × n, so memory and computation can grow roughly quadratically with the number of training samples. Linear SVMs are generally more scalable for very large datasets.

RBF and infinite dimensions — key idea:

The RBF kernel can be mathematically interpreted as an inner product in an infinite-dimensional feature space. The computer does not create that infinite vector; the kernel trick directly computes the equivalent inner product.

Final SVM mental model:

SVM → find a hyperplane → maximize margin → support vectors define the margin → soft margin uses C + hinge loss → kernels allow nonlinear boundaries without explicitly creating high-dimensional features.